

SmartCare-AI

User Manual & High-Level System Roadmap

For Nurses, Administrators, Management and Other Non-Technical Users

Baseline: completed application through F15 End-to-End Acceptance
Backend acceptance: 36 tests / 1,244 assertions
Frontend production build: passed
Repository baseline: main @ 563a027

This manual explains what the system does, how information moves through it, and how staff use the main workflows.

1. What SmartCare-AI Is

SmartCare-AI is a care-management and nursing-home operations system. It connects resident information, daily nursing work, medication administration, clinical monitoring, admissions and discharges, family/facility activities, inventory, billing, reporting, alerts and audit records.

The system uses a Laravel API backend and a React/Vite frontend. AI-related functions support staff with risk signals, predictions, recommendations and alerts, but AI remains advisory and human-authorized.

Who Uses It

User	Typical use
Nurse	Resident care, vitals, medication rounds, care tasks, care records and operational workflows.
Administrator	Operational access plus staff administration, reports, audit history and integrity verification.
Manager / Doctor	Access according to configured role permissions and relevant clinical/management views.
Family / Contacts	Stored against residents; eligible medication-and-meal completions can create family notification records for opted-in contacts.

Operational actions are intended for Active residents. Discharged residents retain their history but are protected from inappropriate new operational records.

2. High-Level Roadmap - How the System Works

Follow the resident journey from arrival to discharge:

Stage	What happens
1. Admit	Save/resume an admission draft, capture resident/family/medical details, complete consent and formal admission.
2. Plan	Assign medications and care plans. Care plans represent recurring needs and generate scheduled care tasks.
3. Daily Care	Nurses use Today and Nurse Tasks, record care, complete medication rounds and confirm linked meals.
4. Monitor	Vitals, weekly checks and monthly glucose build the clinical picture.
5. Respond	Alerts, risk scores, predictions, recommendations and escalation help staff identify attention needs.
6. Communicate	Family contacts/notifications and facility workflows such as visitors, parcels and home leave are tracked.
7. Manage	Inventory, billing, payments, reports, staff administration and audit controls support management.
8. Discharge	Complete discharge, release the room, change the resident to Discharged and retain historical records.

Care Plan -> Care Task -> Care Record

Care Plan = recurring need. Care Task = scheduled occurrence. Care Record = evidence of completion.

3. Main Navigation

Area	Purpose
Today	Read-only operational overview of work requiring attention.
Residents	High-level resident list; open a resident for detailed information.
Medication	Schedules, administration outcomes, meals and history.
Nurse Tasks	Pending, accepted, overdue and completed nursing/care work.
Care Records	Care evidence and unplanned care documentation.
Admissions / Discharges	Resident entry, consent and formal exit workflows.
Inventory	Medicine master, stock and IN/OUT movements.
Monthly Glucose / Weekly Vitals	Routine clinical checks with duplicate-period protection.
Visitors / Home Leave / Parcels	Facility and resident movement/interaction records.
Reports	Administrator management reporting.
AI Intelligence / Clinical Dashboard	Alerts, risk, predictions, recommendations and clinical insight.
Management Dashboard	Administrator-level management overview.

4. Getting Started

- 1 **Sign in** using your assigned staff account.
- 2 **Use the navigation** to open the area needed for your work.
- 3 **Open Residents** for resident-specific work. The list is intentionally simple.
- 4 **Select a resident** to open the detailed profile.
- 5 **Use operational pages** such as Medication, Nurse Tasks, Care Records, Weekly Vitals or Monthly Glucose for daily work.
- 6 **Read success/error messages.** The system prevents many invalid or duplicate actions.
- 7 **Use your own account** so actions are associated with the correct staff member.

Resident Profile

The detailed profile contains Overview, Vitals, Medication, Care, AI Insights, Timeline, Documents and Family information. The resident list remains high-level by design.

5. Admission

The admission interface uses seven steps: Resident, Family, Medical History, Hospital & Medication, Care Assessment, Consent and Review.

- 1 Start or resume a draft. Saving a draft does not create the final resident.
- 2 Enter resident and family/contact information.
- 3 Record medical history and hospitalization information.
- 4 Complete medication and care assessment information as required.
- 5 Complete consent. Core consent is required before formal admission completion; enhanced agreement sections are enforced when used.
- 6 Review and complete admission. The resident becomes Active and the admission date is recorded.

A completed admission is protected from normal draft editing.

6. Daily Care and Nurse Tasks

Care plans describe recurring needs. Generated or manual tasks represent work to do. Care records provide evidence.

Normal workflow: Pending -> Accept Task -> ACKNOWLEDGED -> Complete Task

- 1 Open **Nurse Tasks** and use status/source/priority filters.
- 2 Accept a Pending task.
- 3 After acceptance, choose Complete Task.
- 4 Enter completion notes/evidence.
- 5 Routine-care completion automatically creates the linked Care Record and timeline evidence.
- 6 Completed and Cancelled tasks are historical/read-only.
- 7 For unscheduled care, use **Record Unplanned Care**.

Care-plan frequencies include DAILY, WEEKLY, EVERY_SHIFT and PRN; slots include AM, PM, NIGHT and ANY. PRN plans are not automatically generated.

7. Medication, Meals and Inventory

Slot	Linked meal
AM	Breakfast
PM	Lunch
NIGHT	Dinner
OTHER	No automatic linked meal

- 1 Open Medication or the resident's Medication tab and review the scheduled round.
- 2 Record the outcome: COMPLETED, HELD, REFUSED, UNAVAILABLE, MISSED or DELAYED.
- 3 For non-completed outcomes, provide the required remarks.
- 4 For successful AM/PM/NIGHT completion, confirm the linked meal when appropriate.
- 5 If stock is tracked, successful completion deducts quantity and records an OUT transaction.
- 6 Medication and meal evidence is added to resident history/timeline.
- 7 Eligible completed medication + confirmed meal events can create family notification records for opted-in contacts.

A second final outcome for the same medication occurrence is blocked. A DELAYED occurrence can later be updated to its final outcome.

8. Clinical Monitoring and AI Support

- 1 Record normal vital signs from the resident's Vitals area.
- 2 Use Weekly Vitals for the scheduled weekly check; one check per resident/week is allowed.
- 3 Use Monthly Glucose for the monthly glucose check; one check per resident/month is allowed.
- 4 Review Timeline and clinical/AI views for resulting alerts, risk scores, predictions or recommendations.
- 5 Critical conditions can produce staff notifications and escalation records.

AI is decision support, not an autonomous clinician. Staff remain responsible for reviewing information and taking human-authorized action.

Measurements that were not recorded are not treated as abnormal zero values.

9. Family and Facility Operations

Workflow	Use
Family Contacts	Maintain multiple contacts, relationship, primary/emergency indicators and WhatsApp preference.
Parcels	Record receipt and collection; completed collection is retained historically.
Home Leave	Start leave for an Active resident and record return; conflicting open leave is prevented.
Visitors	Check visitors in and later out; completed records remain historical.
Documents	Manage authenticated resident documents across admission, consent, medical, clinical, medication, care, discharge and other categories.

10. Discharge

- 1 Open Discharges and create a draft for an Active resident.
- 2 Complete discharge date/time, reason and condition at discharge.
- 3 Review the information and complete the discharge.
- 4 The resident becomes Discharged, the discharge date is recorded, the room is released and a timeline event is created.
- 5 Completed discharge records are protected from normal editing/deletion.

Only Active residents can have a new discharge prepared, and only one open discharge draft is allowed per resident.

11. Billing and Payments

- 1 Check whether the resident's monthly fee is configured.
- 2 Billing requires the appropriate completed admission, consent and agreement/payment acknowledgements.
- 3 Generate the monthly invoice for an eligible month; duplicate resident/month invoices are prevented.
- 4 Record payment using an allowed method.
- 5 Partial settlement becomes PARTIALLY_PAID; full settlement becomes PAID.
- 6 Use the receipt information/PDF where required.

Overpayment and payment against already paid/void invoices are blocked.

12. Reports, Staff Administration and Audit

Administrator reporting covers overview, resident census, care, medication, clinical, inventory, billing and family/facility information. Access requires authentication, an Active account and the Administrator role.

Staff Administration

- 1 Administrators can create staff with approved roles.
- 2 Administrators can update staff profiles/status.
- 3 Deactivation revokes the affected staff member's existing tokens.
- 4 Password reset revokes existing tokens.
- 5 An administrator cannot deactivate their own account or remove their own Administrator role through these controls.

Audit and Integrity

Protected activities are recorded in the activity log. New protected audit records participate in a SHA-256 integrity chain. Administrators can review safe audit history and run integrity verification. Signed checkpoint files provide an additional retained verification point.

Ordinary users should never move, edit or delete audit checkpoint/signing material.

13. Common Daily Scenarios

Situation	What to do
Start of shift	Sign in -> Today -> Nurse Tasks -> review overdue/high-priority work -> begin care/medication rounds.
Scheduled care	Nurse Tasks -> Accept -> Complete -> enter evidence -> review resident Care History/Timeline.
AM medication	Medication -> AM round -> record outcome -> confirm breakfast when appropriate -> stock/history/eligible family notification update.
Weekly check	Weekly Vitals -> resident -> enter measurements -> save -> review clinical/AI information.
Temporary leave	Home Leave -> start leave -> record return when resident comes back.
Permanent discharge	Discharges -> prepare -> complete required details -> complete discharge.
Management review	Reports / Management Dashboard -> select period -> review operational and billing information.

14. Development Roadmap - What Was Built

SmartCare-AI was not created only during F1-F15. A substantial backend/frontend foundation existed before the formal phase roadmap. The available project records confirm the progression below. They do **not** preserve a reliable numbered Step 1-Step 70 history, so this document does not invent those step numbers.

Layer	Major outcomes
Earlier foundation	Laravel API structure, authentication/roles, residents, rooms, medications, inventory, vitals, nurse tasks, AI alerts/risk/prediction/recommendation services, dashboards and supporting database structures.
Frontend foundation	Responsive layout/sidebar, Residents/Profile, vitals, medication, care, family, documents and operational navigation.
Operational digitization	Care records, contacts, documents, seven-step admissions, inventory, discharge, monthly glucose, parcels, home leave, visitors, weekly vitals and resident timeline.
Daily Care Operations	Care-plan/task/record model, generation, nurse queue, due/overdue activities, completion evidence, Today integration and QA.
F6-F10	Medication automation, family communication, inventory automation, billing/fees and reports.
F11-F13	Notifications/escalations, staff/user administration, audit/compliance/integrity and signed checkpoints.
F14-F15	Production readiness, migration/deployment hardening and end-to-end acceptance.

Current status: application development is complete through F15. The next stage is controlled client production deployment, not a rewrite.

15. Quick Reference

Need to...	Go to...
Find a resident	Residents
See resident details	Residents -> Resident Profile
See today's work	Today
Accept/complete care	Nurse Tasks
Record unscheduled care	Care Records / Record Unplanned Care
Give medication / confirm meal	Medication
Record clinical measurements	Vitals / Weekly Vitals / Monthly Glucose
Manage stock	Inventory
Admit / discharge	Admissions / Discharges
Visitor / leave / parcel	Visitors / Home Leave / Parcels
Management information	Reports / Management Dashboard
Manage staff / audit	Administrator staff and audit functions

Document scope: This manual reflects the completed SmartCare-AI implementation and project checkpoints available through the final F15 acceptance baseline. Exact branding or labels may be adjusted during client deployment.